Enterprise Network Defense: FortiGate Implementation

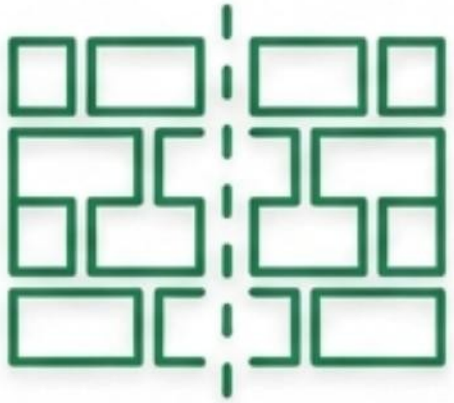
A Zero-Trust & Next-Gen Firewall Architecture Project

FıratCan Bekar | Network Security Architect Portfolio

Project Architecture & Competencies

Secure Access

SSL-VPN Tunnels & User Authentication



Segmentation

Zero-Trust Isolation for Linux (SSH/ICMP) & Windows (RDP)

Traffic Control

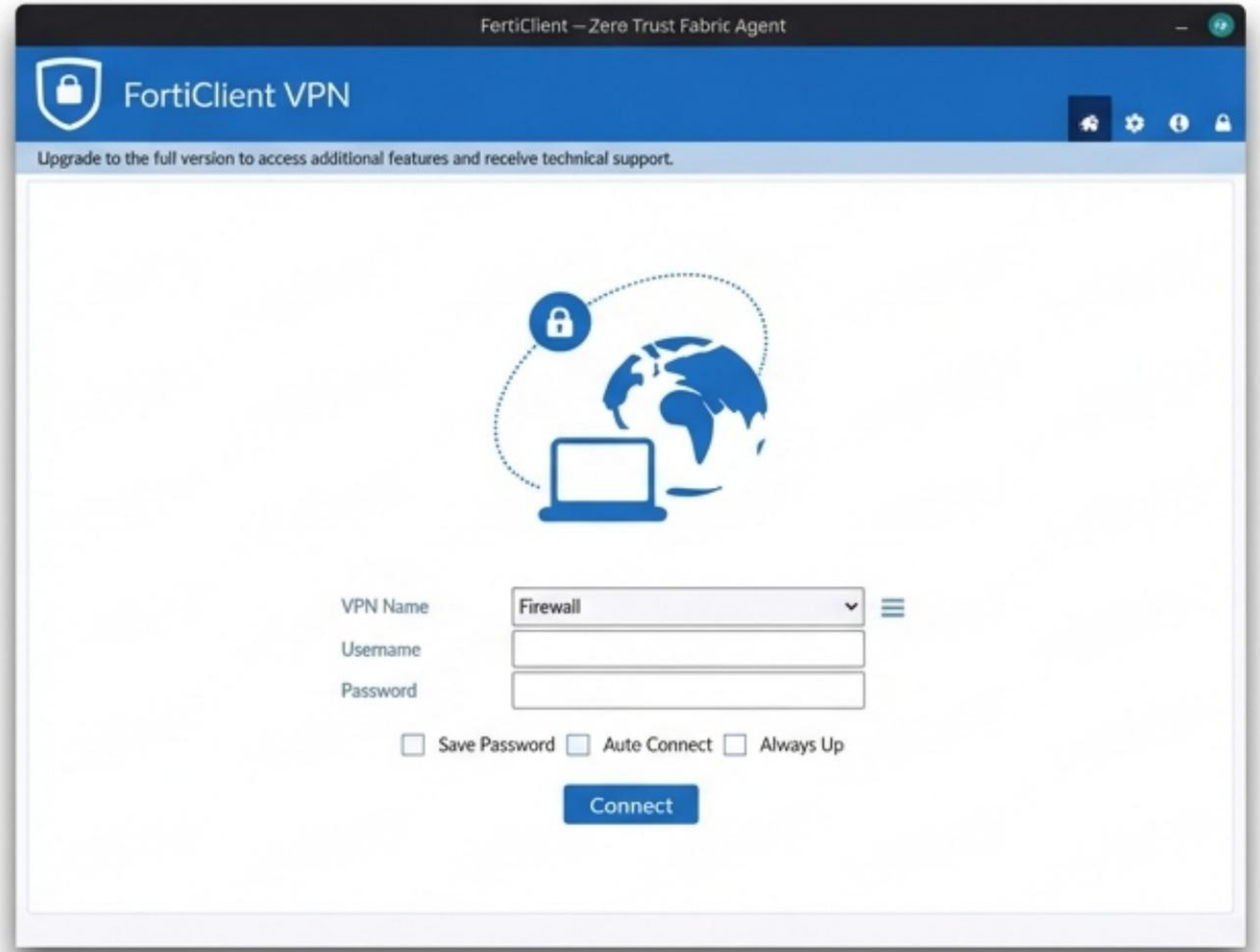
Virtual IPs, Port Forwarding & Layer 7 Application Filtering

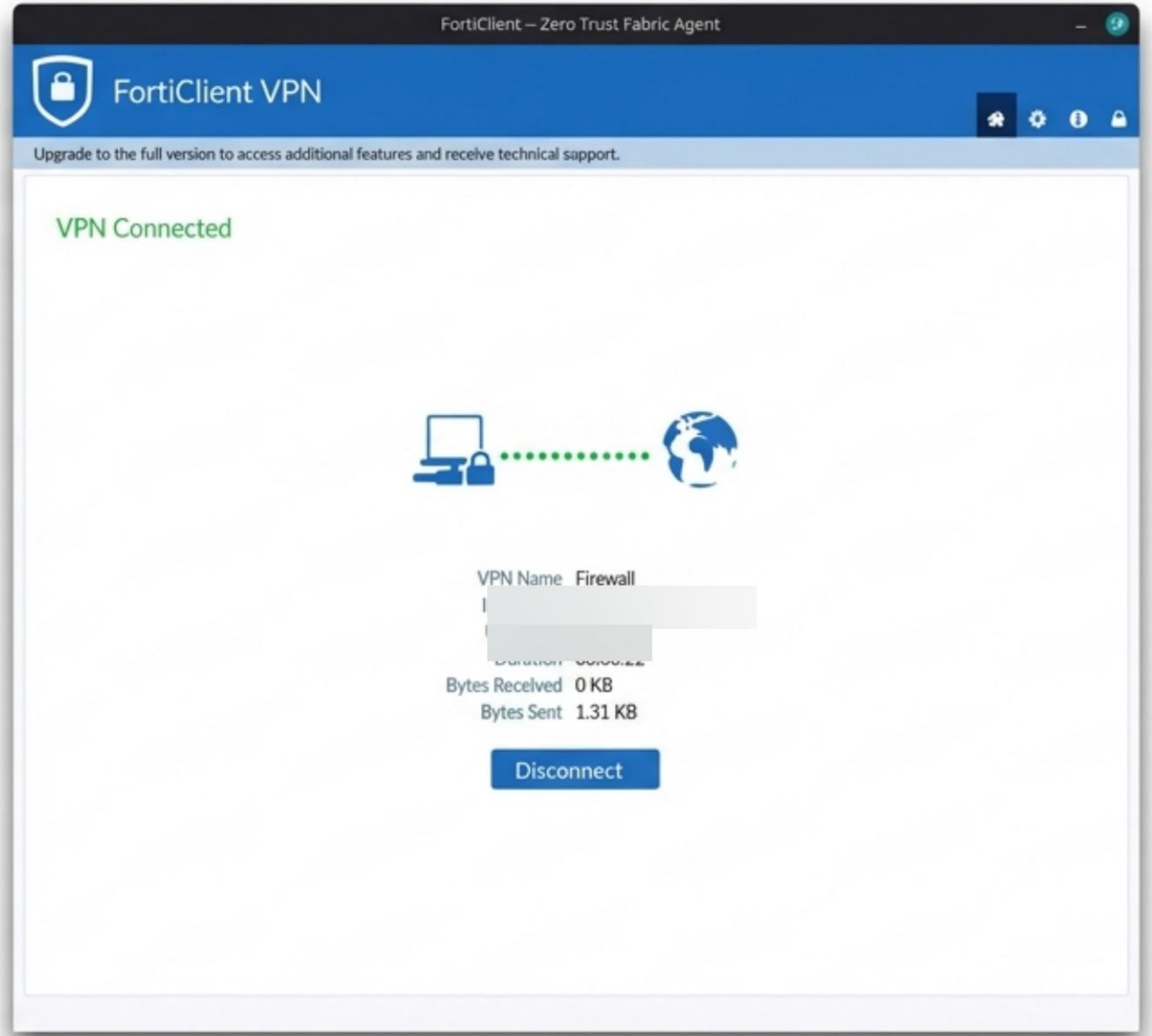
Threat Prevention

IPS Signatures, Antivirus Inspection & DNS Sinkholing

Perimeter Security: SSL-VPN Configuration

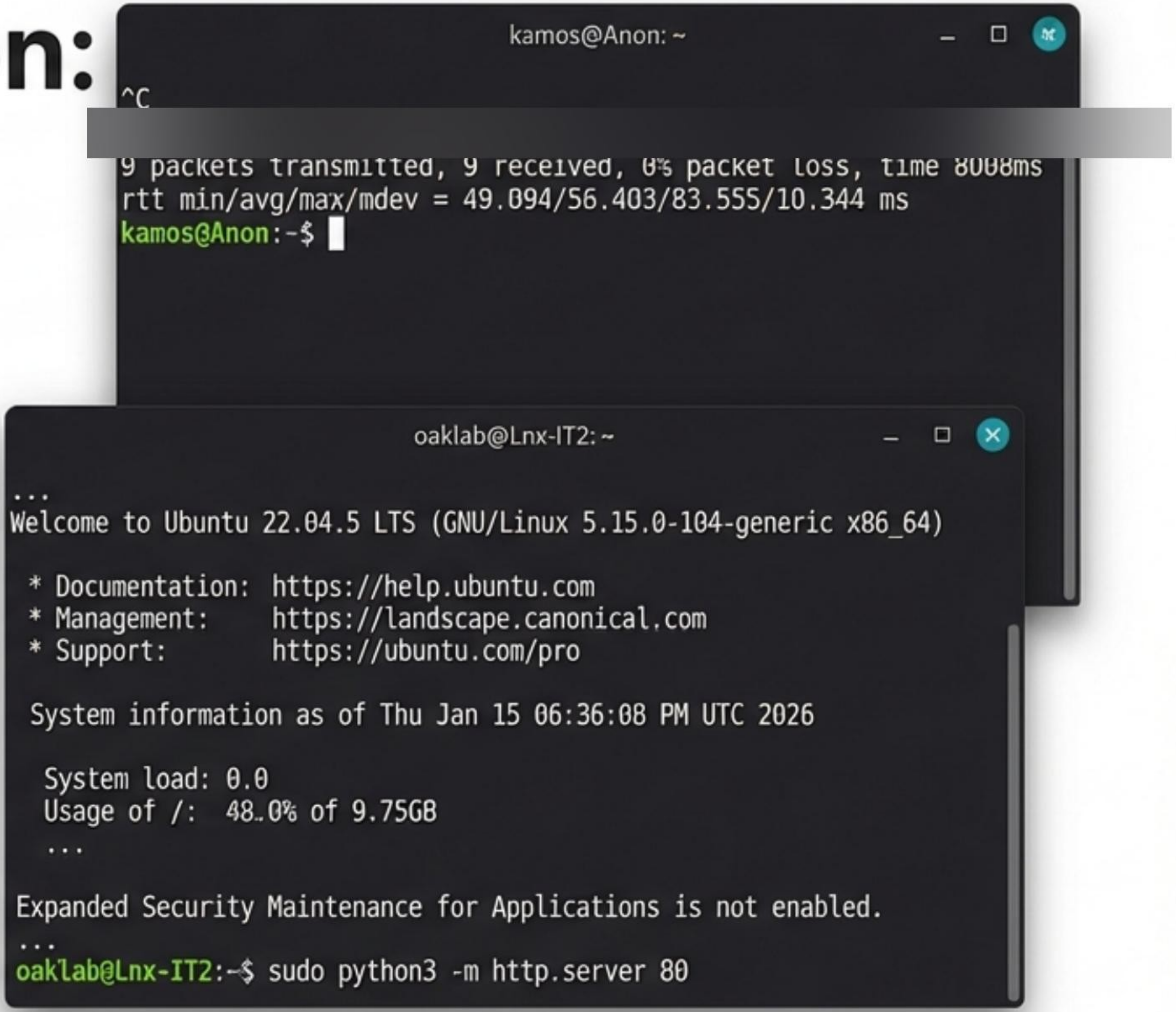
Remote access is secured via FortiClient VPN, establishing an encrypted tunnel into the network fabric.





Internal Segmentation: Linux Environment.

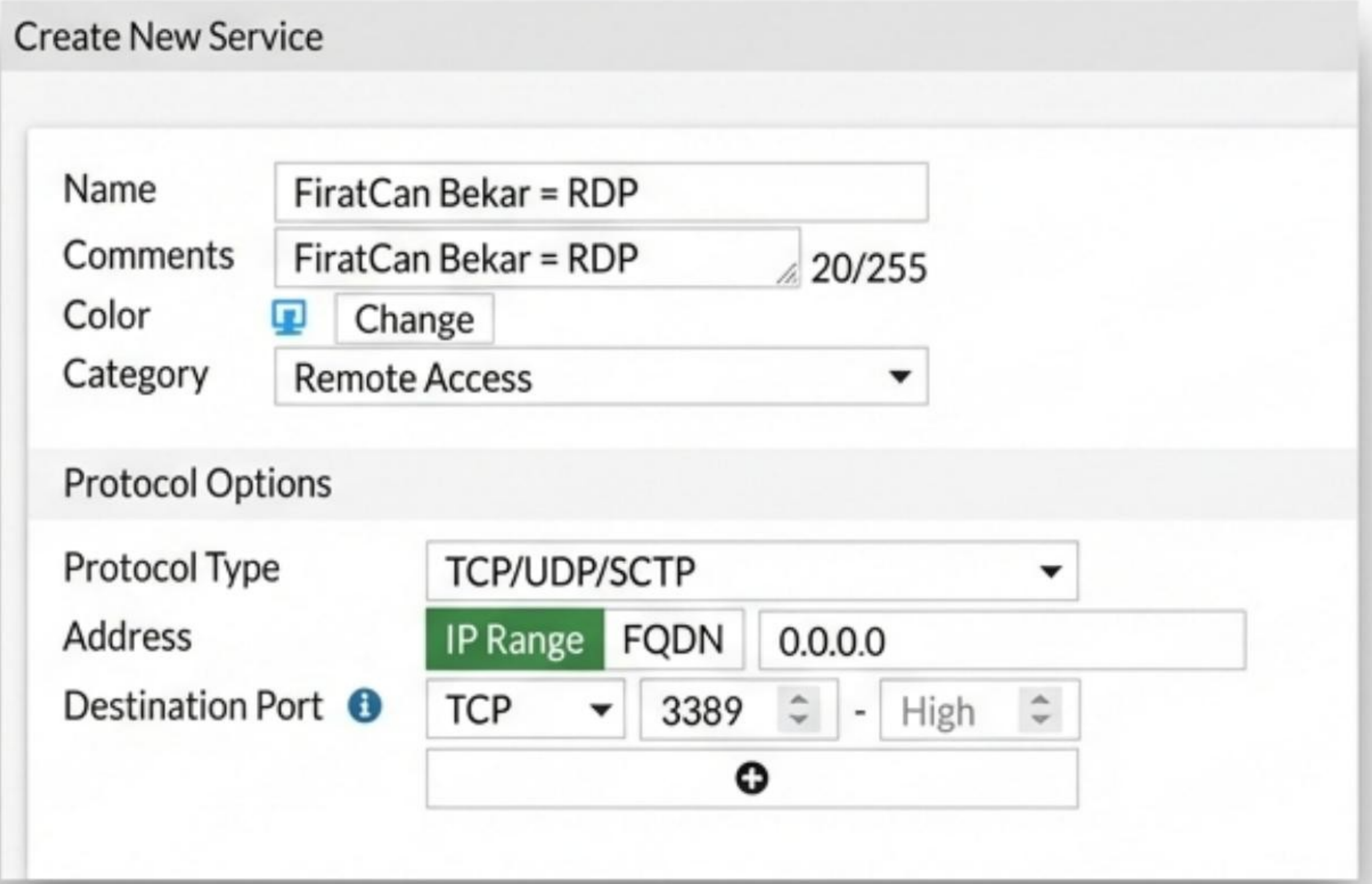
Adhering to Zero-Trust principles, lateral movement is restrictd is restricted. Policy explicitly permits only ICMP (ping) for diagnostics and SSH (Port 22) for management.

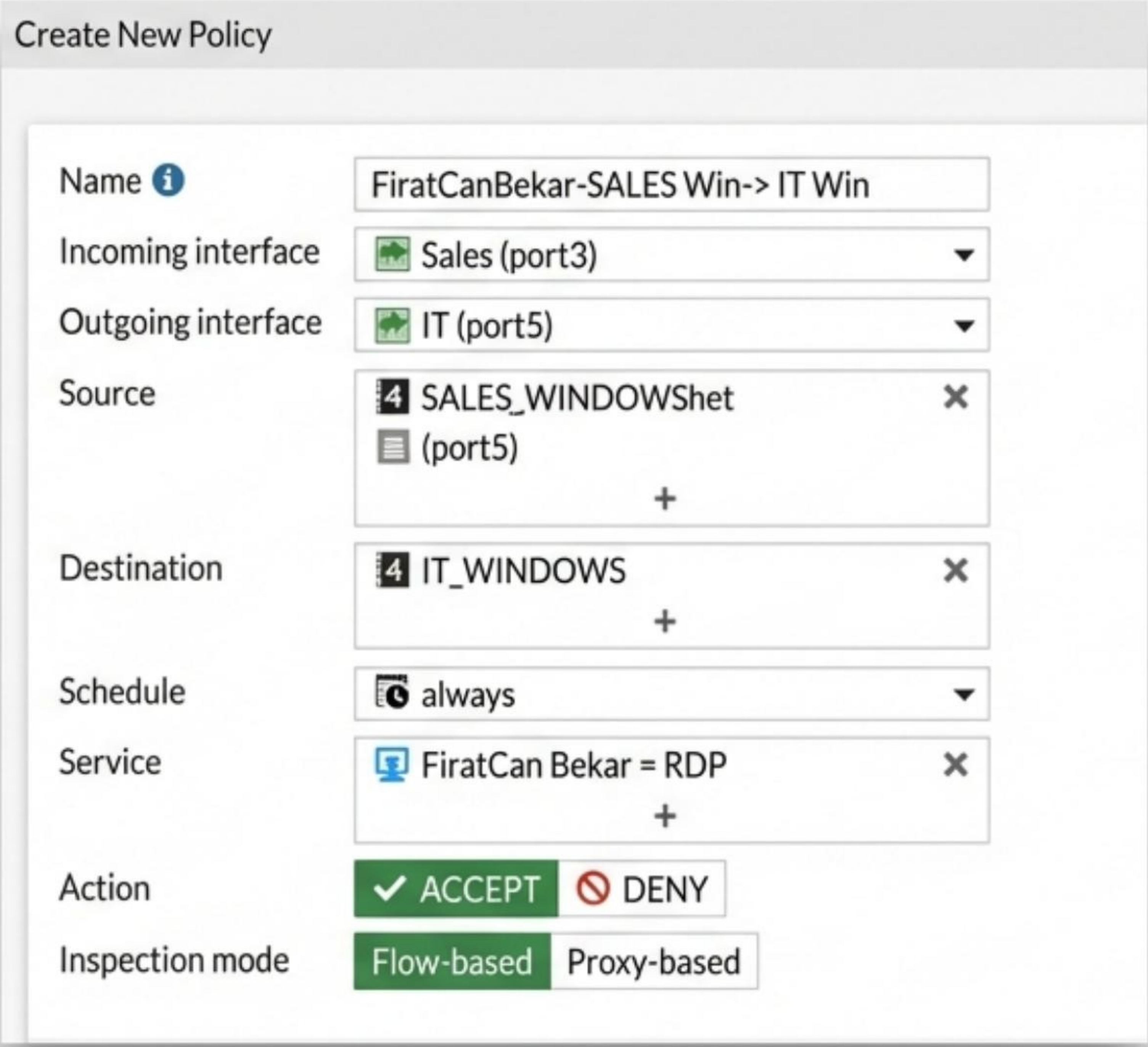


Verified connectivity to Ubuntu 22.04 LTS host.

Protocol Isolation: Windows RDP Control

Windows environments are isolated with strict access controls. Remote Desktop Protocol (TCP/3389) is governed by a dedicated firewall service object.

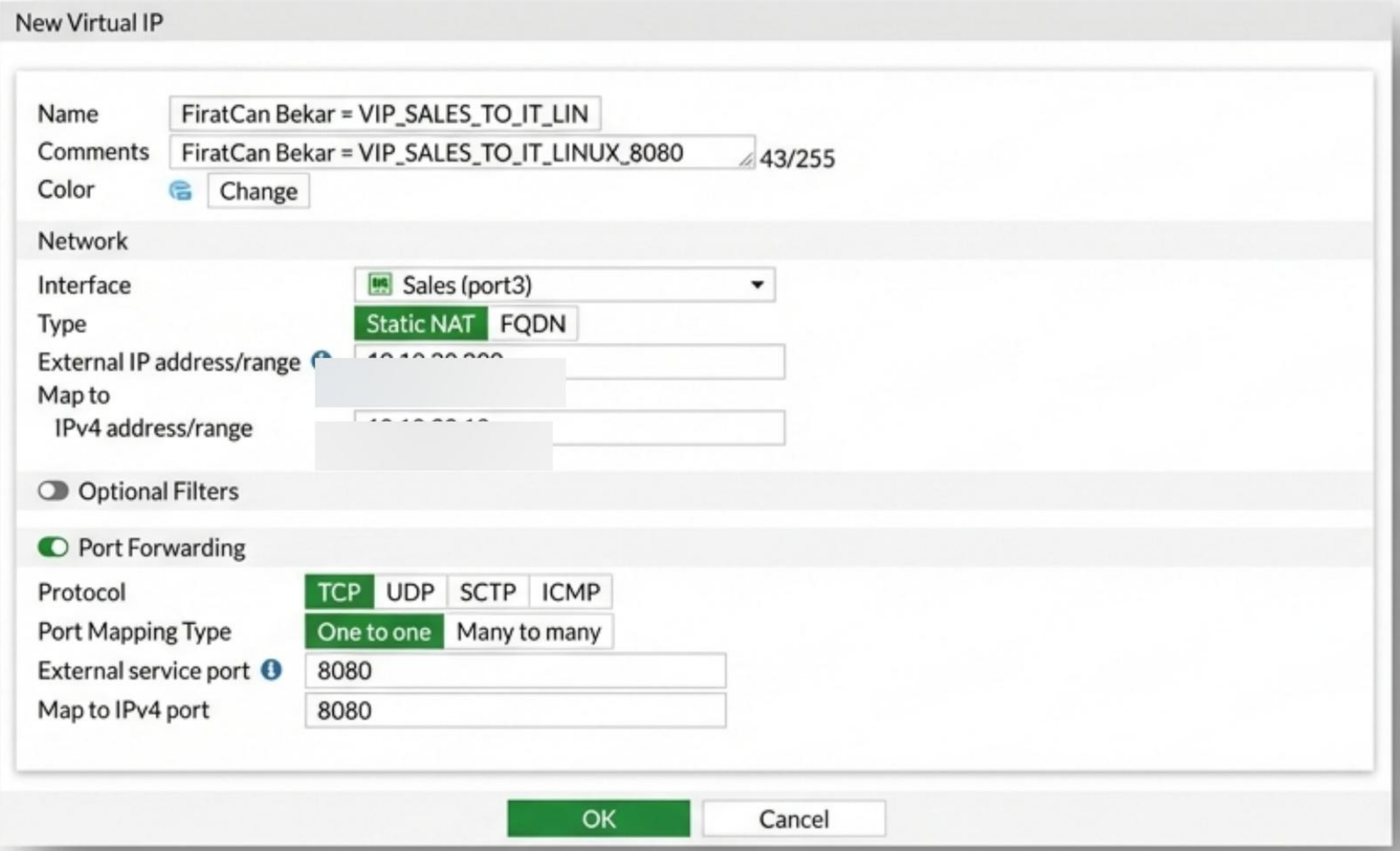




NotebookLM

Inbound Traffic: Virtual IP (VIP) & Port Forwarding.

External access to internal web resources is managed via Static NAT.



Service Verification: Python HTTP Server

To validate the VIP, a Python http.server was deployed. External requests were successfully forwarded to the internal application.

```
oaklab@Lnx-IT2:~$ sudo python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [15/Jan/2026 20:45:58] "GET / HTTP/1.1" 200 -
```

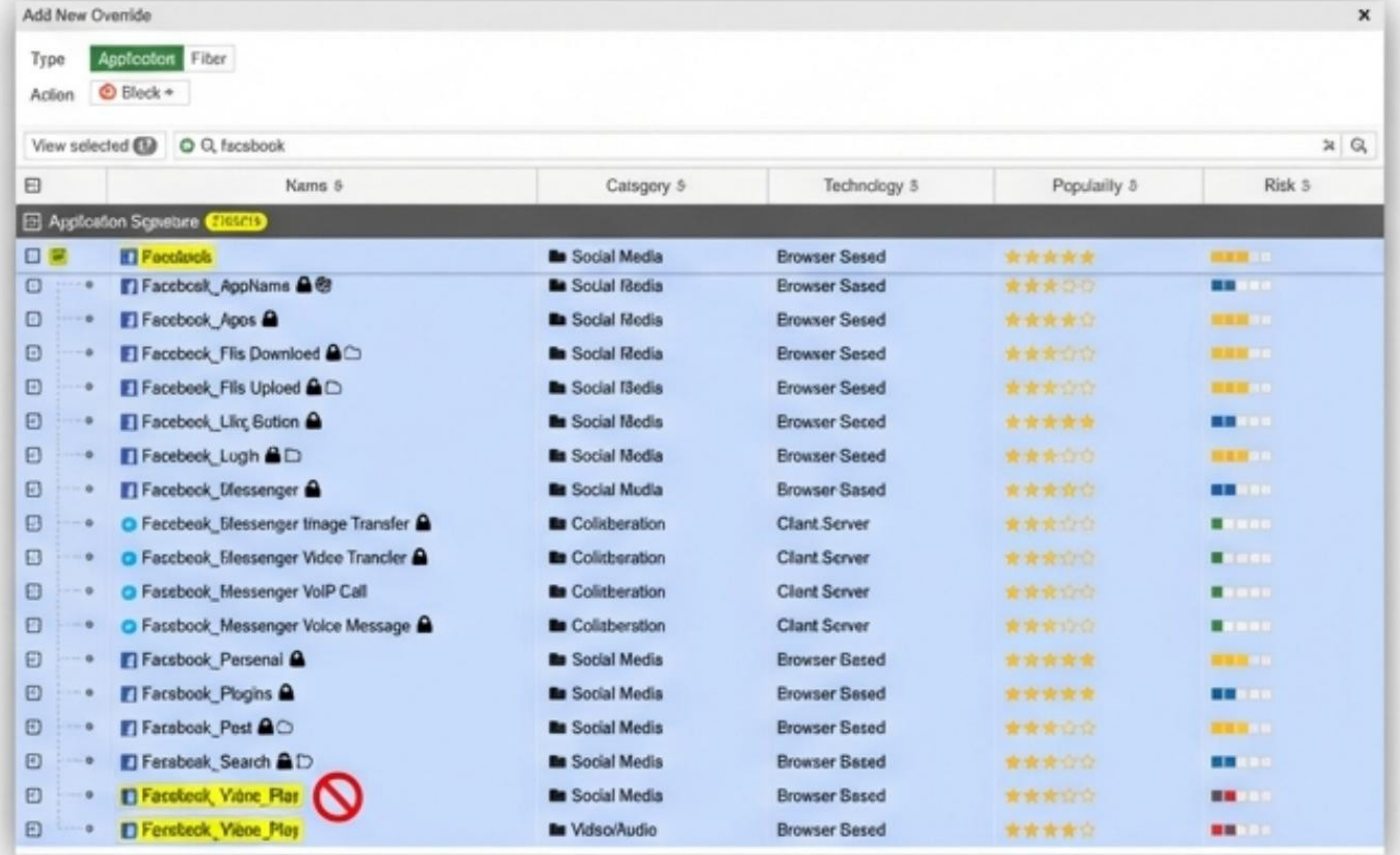
Traffic Flow

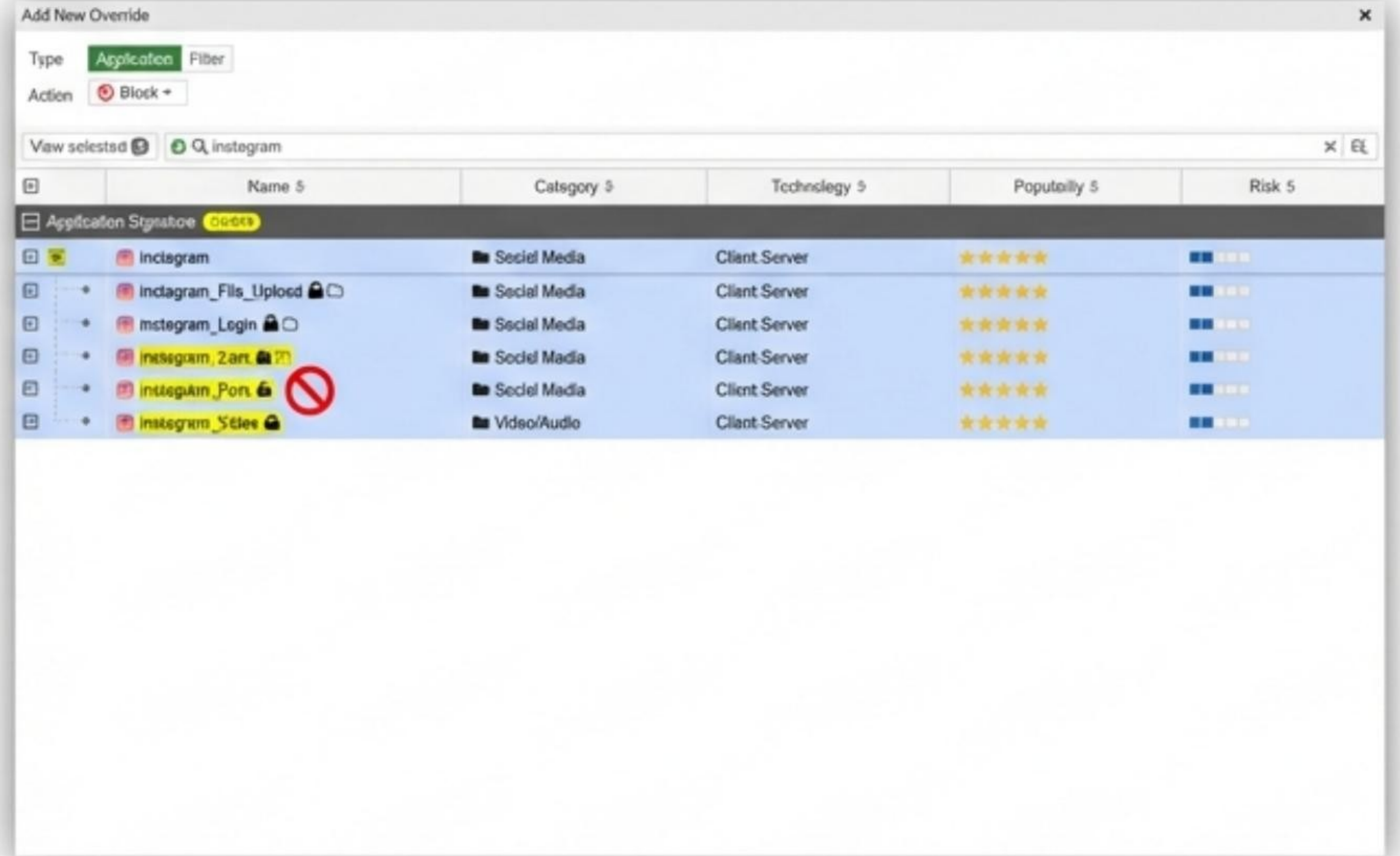
[redacted]

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
    <title>Directory listing for /</title>
  </head>
  <body>
    <h1>Directory listing for /</h1>
    <hr>
    <ul>
     <li><a href=".bash_history">.bash_history</a></li>
     <li><a href=".bash_logout">.bash_logout</a></li>
     <li><a href=".bashrc">.bashrc</a></li>
     <li><a href=".cache/">.cache/</a></li>
     <li><a href=".profile">.profile</a></li>
     <li><a href=".ssh/">.ssh/</a></li>
     <li><a href=".sudo_as_admin_successful">.sudo_as_admin_successful</a></li>
    </ul>
    <hr>
  </body>
</html>
kamoe@Anon:~$
```

Layer 7 Application Control: Social Media.

Deep Packet Inspection (DPI) identifies traffic at the application layer. A rigid block policy is enforced for productivity-draining applications.

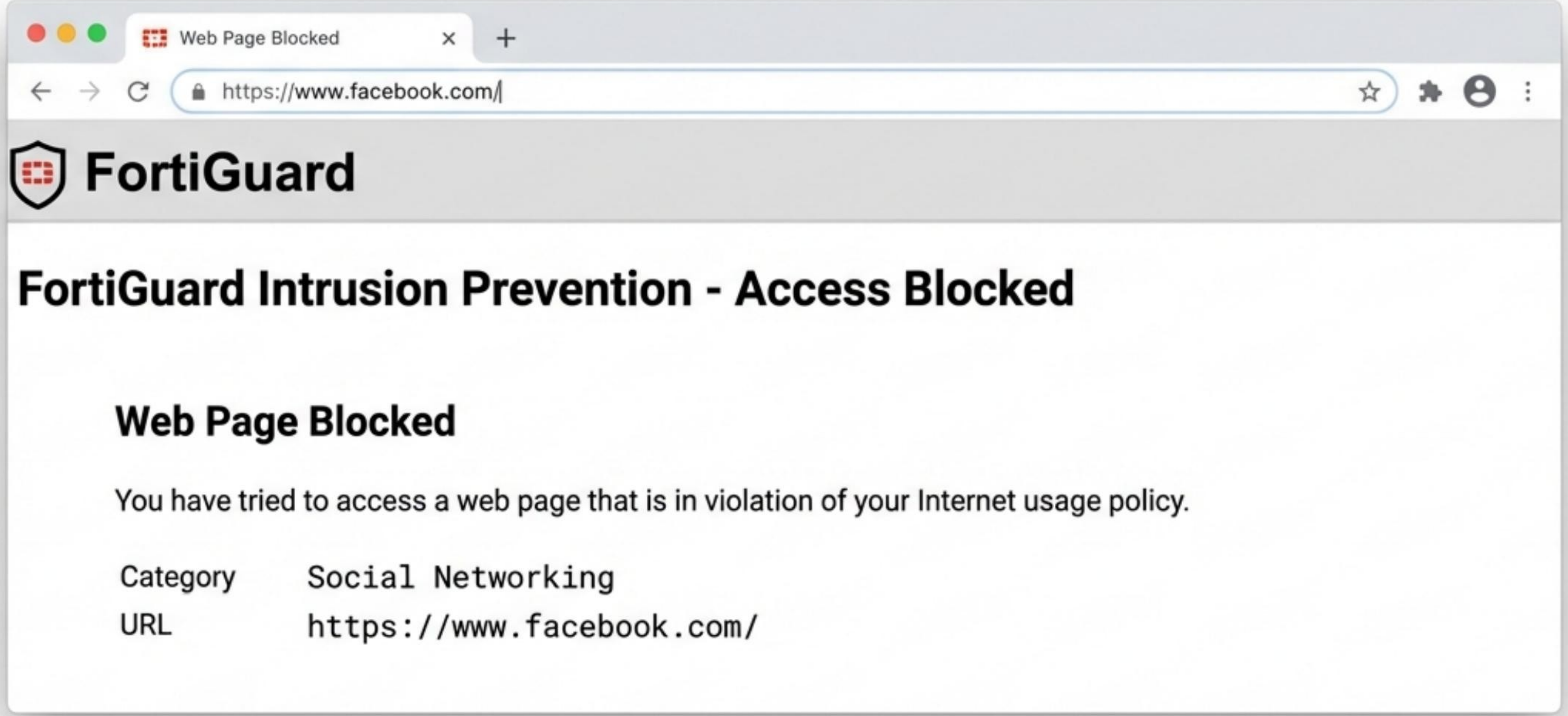




Signatures cover file uploads, login attempts, and video playback.

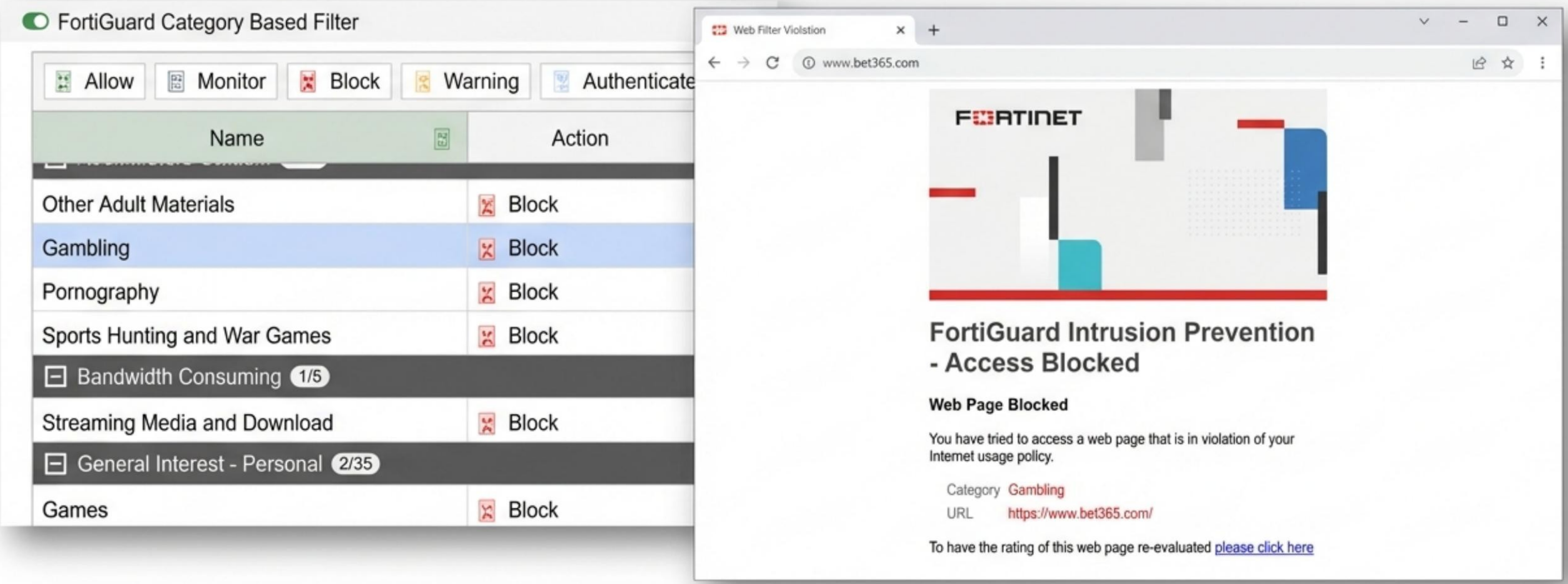
Policy Enforcement: User Experience

When a user attempts to access restricted social platforms, the connection is terminated immediately with a clear violation notice.



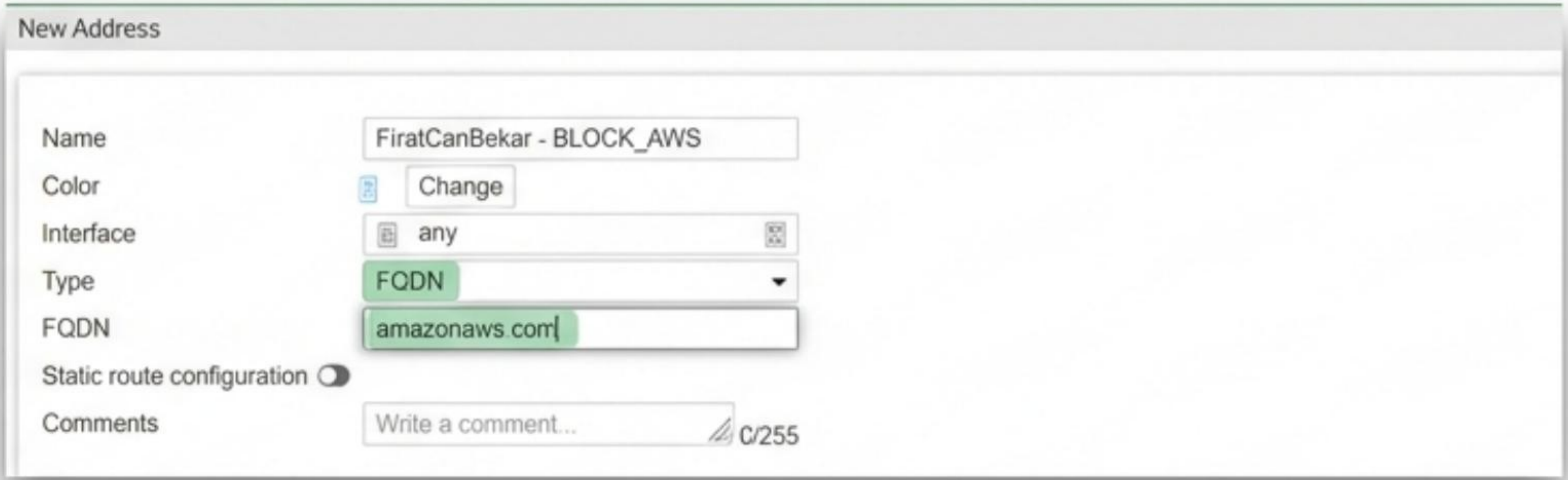
Corporate Web Filtering

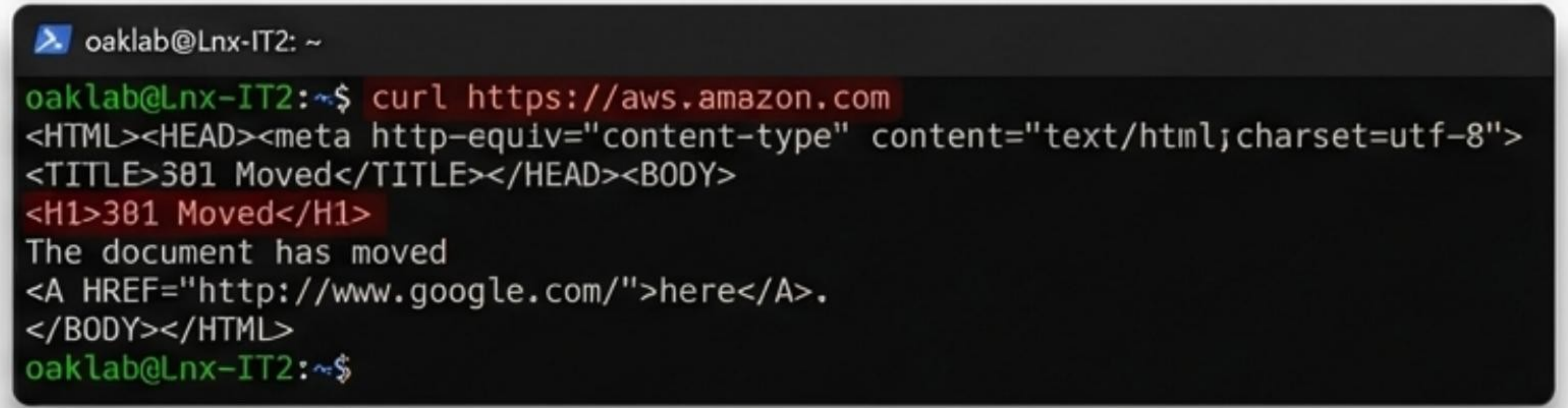
Comprehensive URL filtering sanitizes outbound traffic to reduce liability and optimize network throughput.



Infrastructure Blocking: AWS Cloud Restrictions.

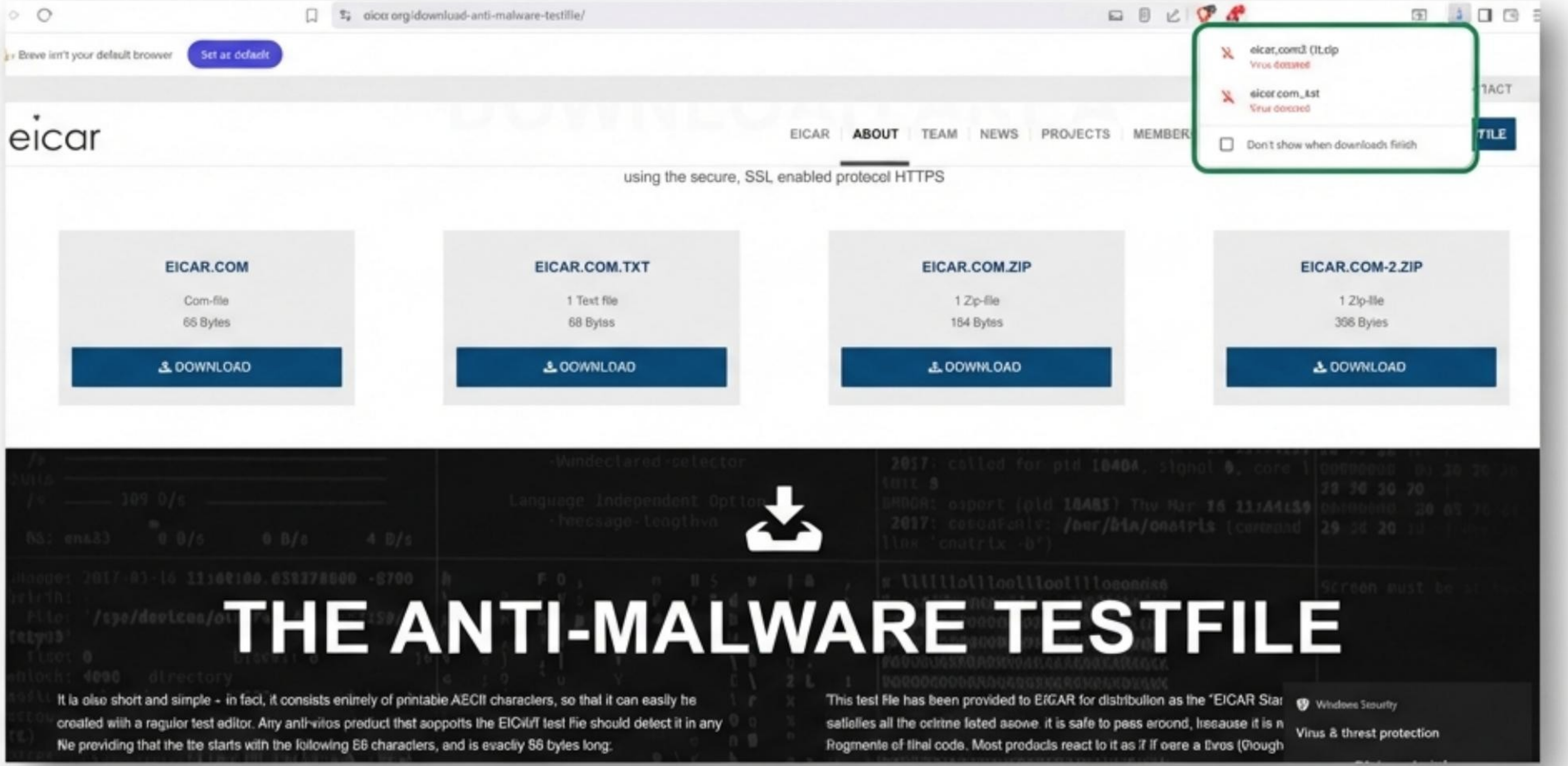
Traffic to specific cloud infrastructure is restricted using FQDN address objects to prevent unauthorized data exfiltration.





Active Threat Protection: Antivirus.

The FortiGate Antivirus profile actively scans HTTP/FTP streams. Successful detection of the EICAR test file confirms payload inspection is active.



Intrusion Prevention System (IPS)

Network-based virtual patching is applied via IPS sensors to drop packets containing known exploit signatures.

IPS Signatures

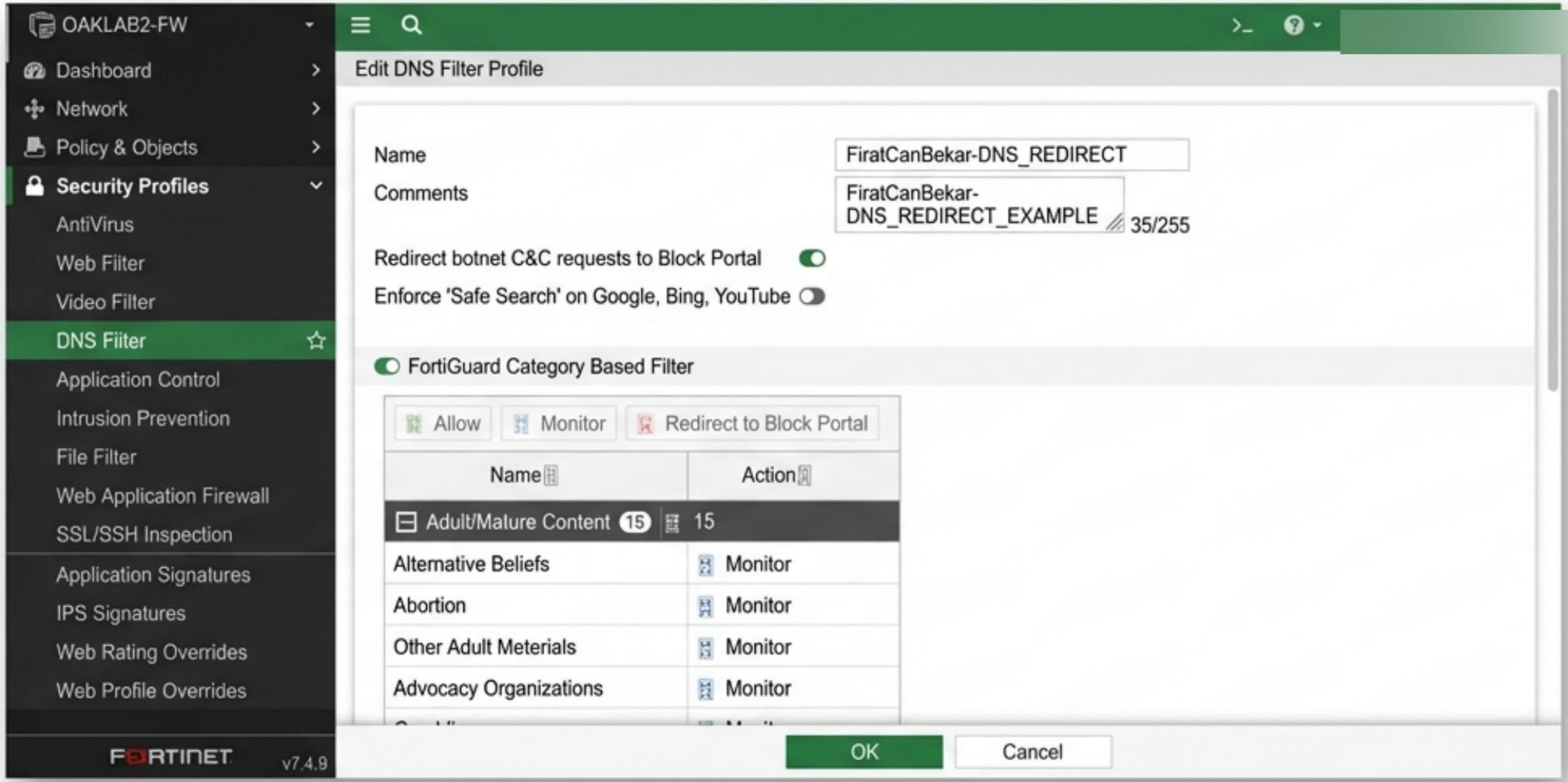
Search

Search

	IPS Signature	IPS Signature	Action
☐	1Panel.image_repo.Create.Command.Injection 1Panel.image_repo.Create.Command.Injection	CVE-2023-36457	Block
☐	3Com.3CDaemon.FTP.Server.Buffer.Overflow 3Com.3CDaemon.FTP.Server.Buffer.Overflow	CVE-2005-0277	Block
☐	3Com.3CDaemon.FTP.Server.Buffer.Overflow 3Com.3CDaemon.FTP.Server.Buffer.Overflow	CVE-2005-0277	Block
☐	3Com.3CDaemon.FTP.Server.Buffer.Overflow 3Com.3CDaemon.FTP.Server.Buffer.Overflow	CVE-2005-0258	Block
☐	3Com.3CDaemon.FTP.Server.Buffer.Overflow	CVE-2005-0367	Block

DNS Security: Sinkhole Configuration.

To identify compromised internal hosts, the DNS filter intercepts requests to known botnet domains and redirects them to a monitored Sinkhole IP.



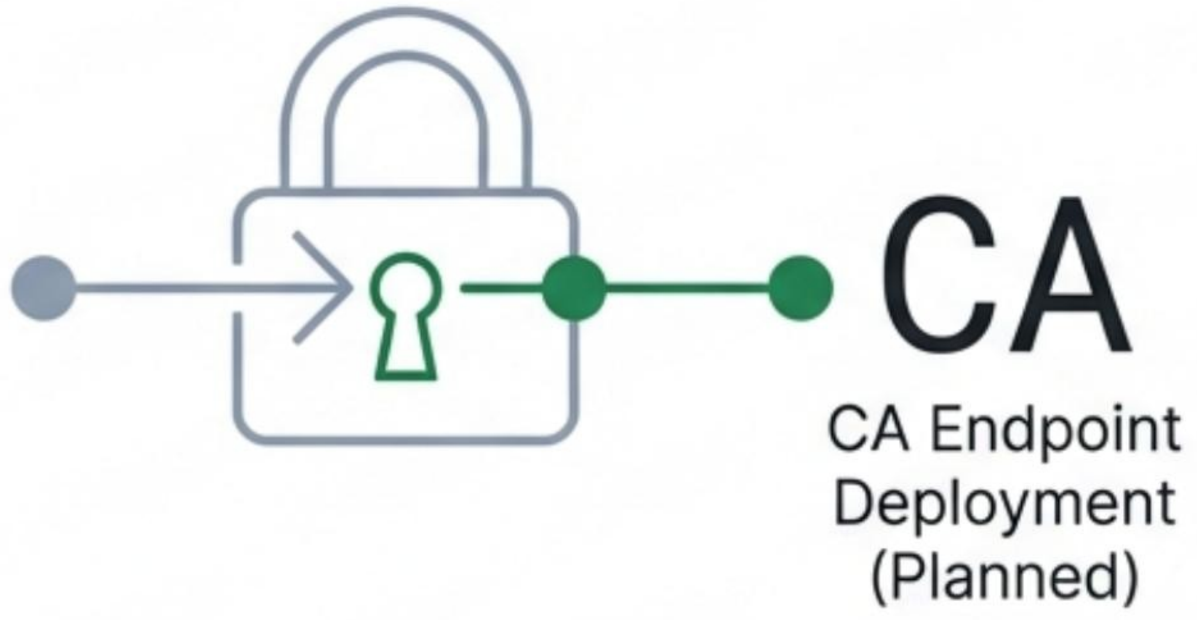
Visibility & Logging

Comprehensive logging is enabled on all firewall policies to ensure auditability. **Every allowed connection and blocked violation** is recorded.

Status	Service	Raw Content
(200 OK)	HTTP	HTTP/1.1 200 OK Accept-Ranges: bytes Content-Length: 703 Content-Type: text/html Date: Sat, 17 Jan 2026 11:28:25 GNT ETag: "b8197ad8a087dc1:0" ETag: "b8197ad8a087dc1:0" Last-Modified: Sat, 17 Jan 2026 11:03:01 GMT Serve...

Technical Constraints & SSL Inspection

In this lab environment, `Deep SSL Inspection` was **partially implemented**. While firewall logs successfully detected application signatures, full content blocking on HTTPS pages relies on `Certificate Authority (CA)` deployment on endpoints, which is a planned future enhancement.



Project Status: Completed.

- ☑ SSL-VPN Remote Access
- ☑ **Zero-Trust** Segmentation (Linux/Windows)
- ☑ Virtual IP & Port Forwarding
- ☑ Layer 7 Application Control
- ☑ Corporate Web Filtering
- ☑ AWS/Cloud FQDN Blocking
- ☑ Antivirus & IPS Deployment
- ☑ DNS Sinkhole Setup